Software Design Description 011

LoRa Bench Test Protocol & Debug

Overview

This document describes the design of the LoRa bench test framework used to validate the RF link between the **ESP32-DevKitV1 (TX master)** and the **ESP32-S3 (echo slave)** prior to integrating the full coefficient packet pipeline.

It also documents four bugs responsible for a **100% packet loss regression**, their root causes, and the corrected design.

> **Hardware:** REYAX RYLR998 x 2 | 915 MHz ISM band | AT-command over UART | Half-duplex LoRa

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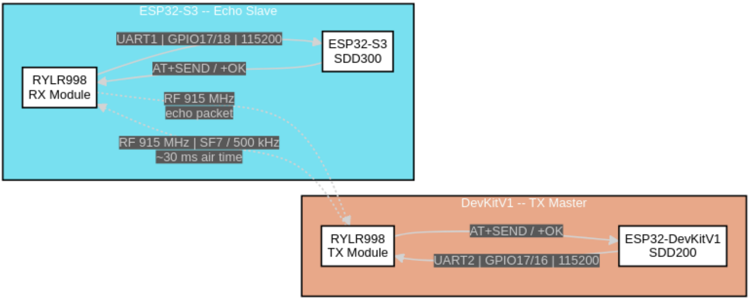
System Architecture

The bench test isolates the wireless link between the two ESP32 boards. All other subsystems (camera, FPGA, sensors, WiFi) are disabled by the `TEST_MODE_LORA_BENCH` build flag.



ESP32-S3 -- Echo Slave
UART1 | GPIO17/18 | 115200
ESP32-S3 SDD300
RYLR998 RX Module
AT+SEND / +OK
RF 915 MHz echo packet
RF 915 MHz | SF7 / 500 kHz ~30 ms air time
DevKitV1 -- TX Master
RYLR998 TX Module
AT+SEND / +OK
ESP32-DevKitV1 SDD200
UART2 | GPIO17/16 | 115200

Hardware Setup

DevKitV1 UART -> RYLR998 TX

Signal	ESP32 Pin	RYLR998 Pin	Direction
UART TX	GPIO17	RX	ESP32 -> Module
UART RX	GPIO16	TX	Module -> ESP32
UART Port	UART_NUM_2	–	–
Baud Rate	115200	–	8N1, no flow ctrl

ESP32-S3 UART -> RYLR998 RX

Signal	ESP32 Pin	RYLR998 Pin	Direction
UART TX	GPIO17	RX	ESP32 -> Module

UART RX	GPIO18	TX	Module -> ESP32
UART Port	UART_NUM_1	–	–
Baud Rate	115200	–	8N1, no flow ctrl

Note: The S3 RX pin (GPIO18) differs from the DevKitV1 RX pin (GPIO16). Swapping TX/RX on the S3 silently kills all comms with no error output.

AT Command Protocol

The RYLR998 is not a transparent UART radio – it has its own AT command layer that must be understood to use the module correctly.

```
---------------- TRANSMIT SIDE ----------------

ESP32  -->  AT+SEND=<dest>,<len>,<data>\r\n  -->  RYLR998
ESP32  <--  +OK\r\n                          <--  RYLR998
                (payload buffered; RF TX begins after +OK)

---------------- RECEIVE SIDE -----------------

RYLR998 -->  +RCV=<src>,<len>,<data>,<rssi>,<snr>\r\n  --> ESP32
```

Field definitions

Field	Description
`<dest>`	Destination module address (must match receiver's ADDRESS)
`<src>`	Source module address
`<len>`	Payload byte count (1-240)
`<data>`	Raw payload bytes
`<rssi>`	Received signal strength (dBm, negative)
`<snr>`	Signal-to-noise ratio (dB)

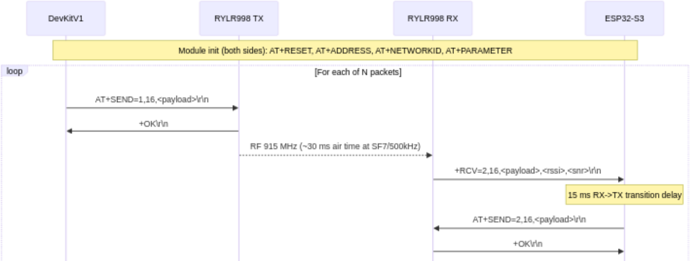
Initialization sequence (both modules)

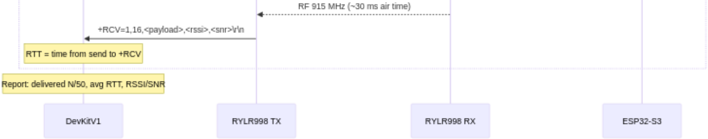
```
AT+RESET           ->  +RESET   (wait 1500 ms after reset)
AT+ADDRESS=<n>     ->  +OK
AT+NETWORKID=<n>   ->  +OK      <- both modules MUST share same ID
AT+PARAMETER=<SF>,<BW>,<CR>,<Preamble>  ->  +OK   <- must match on both
AT+ADDRESS?        ->  +ADDRESS=<n>      (verify)
AT+NETWORKID?      ->  +NETWORKID=<n>    (verify)
AT+PARAMETER?      ->  +PARAMETER=...    (verify)
AT+BAND?           ->  +BAND=...         (verify)
```

Bench Test Design

The bench is a **request-response echo test**. The DevKitV1 is the master; the S3 is the slave. One packet is in-flight at a time.

Complete packet exchange sequence


DevKitV1
RYLR998 TX
RYLR998 RX
ESP32-S3
Module init (both sides): AT+RESET, AT+ADDRESS, AT+NETWORKID, AT+PARAMETER
loop
[For each of N packets]
AT+SEND=1,16,<payload>\r\n
+OK\r\n
RF 915 MHz (~30 ms air time at SF7/500kHz)
+RCV=2,16,<payload>,<rssi>,<snr>\r\n
15 ms RX->TX transition delay
AT+SEND=2,16,<payload>\r\n
+OK\r\n


RF 915 MHz (~30 ms air time)
+RCV=1,16,<payload>,<rssi>,<snr>\r\n
RTT = time from send to +RCV
Report: delivered N/50, avg RTT, RSSI/SNR
DevKitV1
RYLR998 TX
RYLR998 RX
ESP32-S3

Payload format

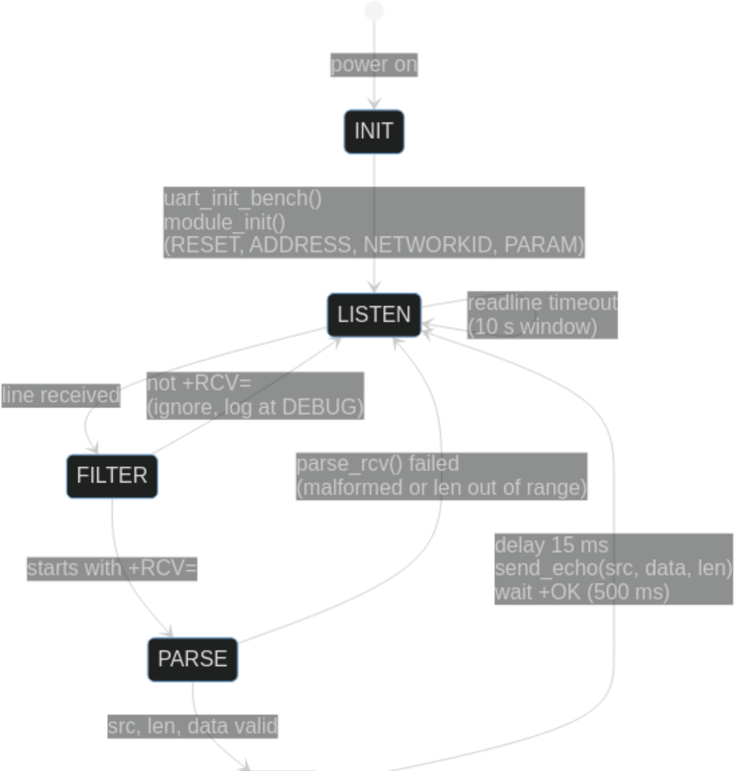
Each packet is `BENCH_PKT_SIZE` bytes (default 16):

```
Bytes [0..7]  : sequence number as 8-character lowercase hex ASCII  e.g. "00000003"
Bytes [8..15] : padding byte 'A' (0x41) repeated
```

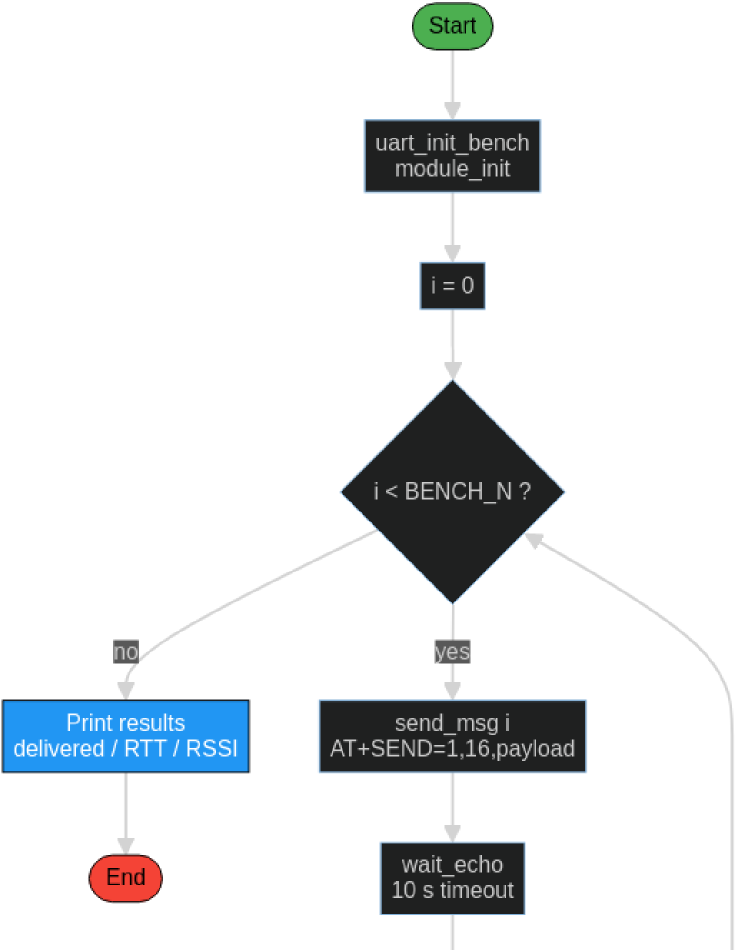
> No null byte is placed in the wire payload. `snprintf` is used into a temporary buffer and then `memcpy`'d without the null terminator.

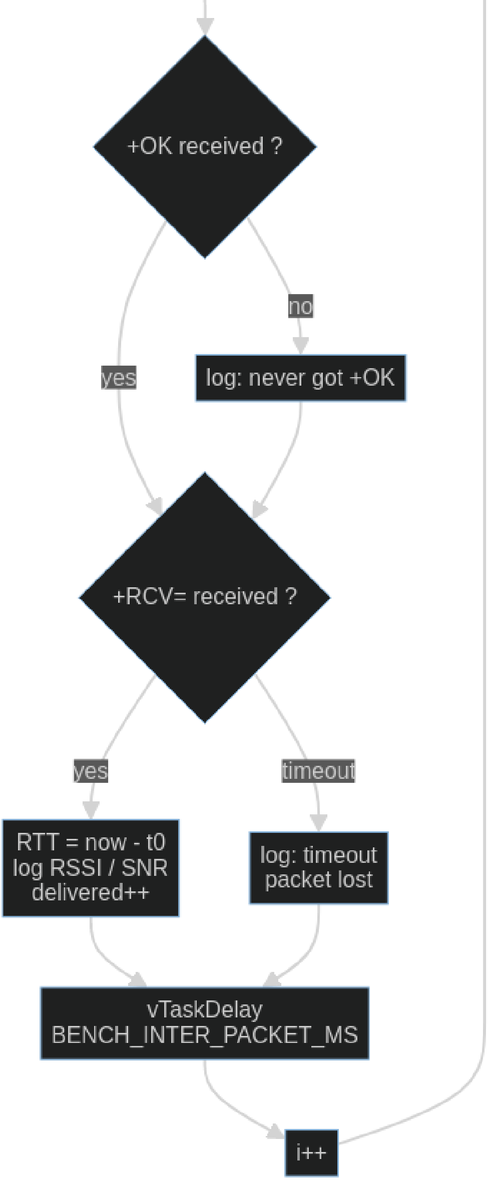
State Machines

S3 Echo Slave


power on
INIT
uart_init_bench()
module_init()
(RESET, ADDRESS, NETWORKID, PARAM)
LISTEN
readline timeout
(10 s window)
line received
not +RCV=
(ignore, log at DEBUG)
FILTER
parse_rcv() failed
(malformed or len out of range)
starts with +RCV=
delay 15 ms
send_echo(src, data, len)
wait +OK (500 ms)
PARSE
src, len, data valid

DevKitV1 TX Master


Start
uart_init_bench
module_init
i = 0
i < BENCH_N ?
no
yes
Print results
delivered / RTT / RSSI
send_msg i
AT+SEND=1,16,payload
End
wait_echo
10 s timeout


+OK received ?
yes
no
log: never got +OK
+RCV= received ?
yes
timeout
RTT = now - t0
log RSSI / SNR
delivered++
log: timeout
packet lost
vTaskDelay
BENCH_INTER_PACKET_MS
i++

Root Cause Analysis

Four bugs were found in the bench code. **Bugs 1 and 4 caused 100% packet loss.** Bugs 2 and 3 produced incorrect metrics.

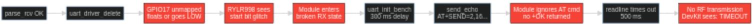
Bug 1 – S3: UART driver delete/reinstall on every echo iteration *(root cause)*

File: `esp/esp32s3_wroom/main/src/lora_bench.c` lines 219-224

Broken code:

```c
uart_driver_delete(LORA_UART_NUM);    // <- unregisters GPIO from UART peripheral
vTaskDelay(pdMS_TO_TICKS(10));
uart_init_bench();                    // <- reinstalls driver + 300 ms settle delay
vTaskDelay(pdMS_TO_TICKS(10));
send_echo(src, data, len);
```

Failure path:


parse_rcv OK
uart_driver_delete
GPIO17 unmapped floats or goes LOW
RYLR998 sees start bit glitch
Module enters broken RX state
uart_init_bench 300 ms delay
send_echo AT+SEND=2,16...
Module ignores AT cmd no +OK returned
readline times out 500 ms
No RF transmission DevKit sees: TIMEOUT

Fix: Remove the reinit entirely. The UART driver should be installed once and left running. The RYLR998 only needs 15 ms to transition from receive to transmit mode.

```c
// REMOVED: uart_driver_delete / uart_init_bench block

vTaskDelay(pdMS_TO_TICKS(15));    // RYLR998 RX->TX mode switch (per README)
send_echo(src, data, len);
```

Bug 2 – DevKitV1: Null byte embedded in wire payload

File: `esp/esp32_devkitv1/main/src/interprocessor/lora_bench.c`

```c
// BROKEN -- snprintf writes '\0' at payload[8]:
memset(payload, 'A', size);
snprintf((char *)payload, 9, "%08lx", (unsigned long)seq);

// FIXED -- separate buffer, memcpy without null:
memset(payload, 'A', size);
char seq_str[9];
snprintf(seq_str, sizeof(seq_str), "%08lx", (unsigned long)seq);
memcpy(payload, seq_str, 8);
```

Some RYLR998 firmware versions treat the `AT+SEND` data field as a C string and truncate at the first `\0`. Sending a 16-byte payload with a null at byte 8 could cause the module to transmit only 8 bytes, causing the echo to arrive with `len=8`. The DevKitV1's RSSI parsing offset (hardcoded to skip `BENCH_PKT_SIZE` bytes) would then land in the wrong field.

Bug 3 – DevKitV1: RTT timestamps 2.4% high

File: `esp/esp32_devkitv1/main/src/interprocessor/lora_bench.c`

```c
// BROKEN -- >> 10 divides by 1024, not 1000:
int64_t t0  = esp_timer_get_time() >> 10;
while ((esp_timer_get_time() >> 10) - t0 < timeout_ms)
int64_t rtt = (esp_timer_get_time() >> 10) - t0;

// FIXED -- true milliseconds:
int64_t t0  = esp_timer_get_time() / 1000;
while ((esp_timer_get_time() / 1000) - t0 < timeout_ms)
int64_t rtt = (esp_timer_get_time() / 1000) - t0;
```

`esp_timer_get_time()` returns microseconds. `>> 10` (/1024) gives units of 1024 us each. All RTT values and the timeout comparison used 1024 us as "1 ms," producing results 2.4% higher than actual and an effective timeout of 10.24 s instead of 10.00 s.

Bug 4 – Startup race: DevKitV1 transmits before S3 is listening

Observed behavior:

DevKitV1 sends all 50 packets, receives `+OK` for each (local module accepted the AT command), but never receives a `+RCV` echo. 100% timeout. Meanwhile the S3 log shows it received only one packet – and that packet was **seq 2**, not seq 0. Packets 0 and 1 were transmitted over RF while the S3 was still in its post-init delay and not yet calling `readline`.

Why this happens:

The RYLR998 `+OK` response is **not a link-layer acknowledgment**. It means only that the local module accepted the `AT+SEND` command and will begin RF transmission. LoRa is a broadcast PHY – there is no pairing, no connection state, and no indication from the module that a remote receiver exists or is ready. The DevKitV1 has no way to know whether the S3 is listening.

Both boards boot independently. Their init sequences (`uart_init_bench` -> 500 ms settle -> `module_init` with AT+RESET -> 1500 ms -> AT config -> 500 ms settle -> `listening...`) take different wall-clock times depending on bootloader, flash size detection, and PSRAM init (S3 only). In the observed run:

```
DevKitV1 sends first packet:  ~3184 ms after boot
S3 reaches listening state:   ~3757 ms after boot
```

The S3 is ~573 ms late. At the bench test's pace (~10.2 s per packet due to timeout + inter-packet delay), this only costs the first 1-2 packets. But those lost packets start the DevKitV1's timeout cascade, and any secondary failure (USB disconnect, module lockup from an unread +RCV in the S3's UART buffer) turns the rest of the trial into 100% loss.

Why a fixed delay is not the right fix:

Adding e.g. `vTaskDelay(pdMS_TO_TICKS(5000))` on the DevKitV1 before the send loop would paper over the race for bench-distance testing, but:

1. It's fragile – the S3 boot time depends on flash size, PSRAM configuration, and any future init work. A change to either board's startup could reintroduce the race.
2. It doesn't scale – when this protocol moves off the bench and onto hardware with variable power-on sequencing (battery, solar, field deployment), there's no guarantee which board powers up first.
3. It masks real failures – if the S3 fails to init its module entirely, the DevKitV1 will silently wait out the delay and then run a full 50-packet trial against a dead link.

Fix: Application-level ready handshake

Since the RYLR998 provides no link-layer readiness signal, the application must implement one. The protocol is:

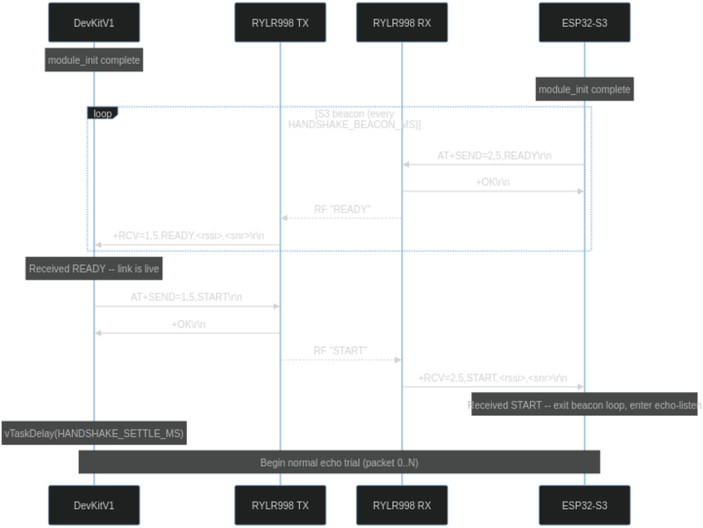
1. **S3 (echo slave)** completes `module_init`, then enters a beacon loop: it sends a short `READY` packet to the DevKitV1's address every `HANDSHAKE_BEACON_MS` (e.g., 1000 ms) and listens for a `START` response between beacons.

2. **DevKitV1 (TX master)** completes `module_init`, then enters a handshake-wait loop: it calls `readline` with a short timeout, looking for a `+RCV` containing the `READY` payload. Upon receiving it, it sends a `START` packet back to the S3's address.

3. **S3** receives the `START` packet (via `+RCV` during its listen window between beacons), exits the beacon loop, and enters the normal echo-listen state.

4. **DevKitV1** waits a short settle time after sending `START` (so the S3 has time to transition to listen mode), then begins the 50-packet trial.

If either board is not yet ready, the handshake simply hasn't completed – there is no timeout cascade, no wasted packets, and the trial only starts when both sides have confirmed link connectivity.


DevKitV1
RYLR998 TX
RYLR998 RX
ESP32-S3
module_init complete
module_init complete
loop
[S3 beacon (every HANDSHAKE_BEACON_MS)]
AT+SEND=2,5,READY\r\n
+OK\r\n
RF "READY"
+RCV=1,5,READY,<rssi>,<snr>\r\n
Received READY -- link is live
AT+SEND=1,5,START\r\n
+OK\r\n
RF "START"
+RCV=2,5,START,<rssi>,<snr>\r\n
Received START -- exit beacon loop, enter echo-listen
vTaskDelay(HANDSHAKE_SETTLE_MS)
Begin normal echo trial (packet 0..N)
DevKitV1
RYLR998 TX
RYLR998 RX
ESP32-S3

Handshake payload format

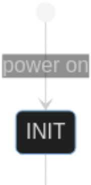
Packet	Sender	Payload	Length
READY	S3	`READY`	5
START	DevKitV1	`START`	5

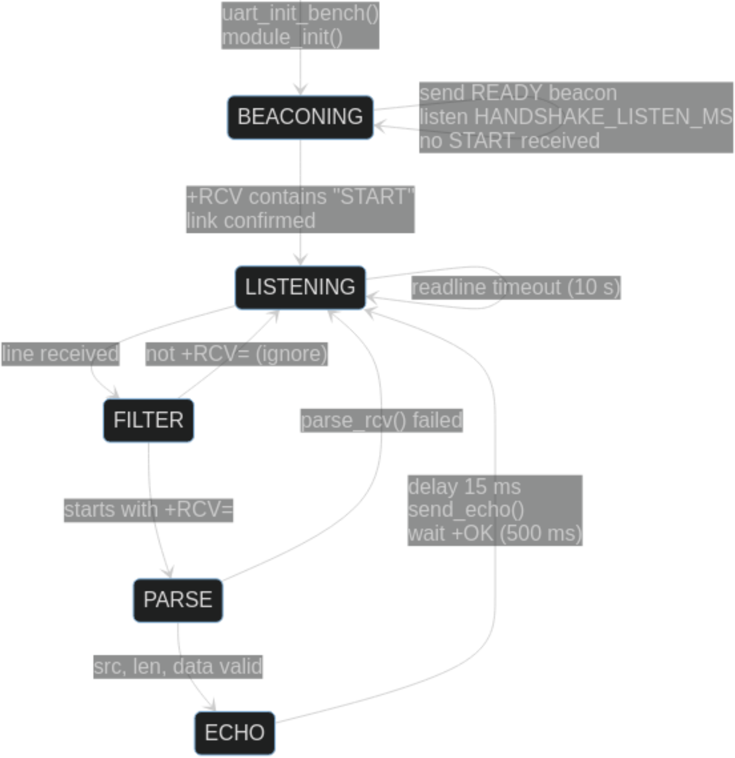
These are plain ASCII, no sequence number or framing needed. The `parse_rcv` function already extracts the data field – the handshake just checks `memcmp(data, "READY", 5)` or `memcmp(data, "START", 5)`.

Handshake configuration

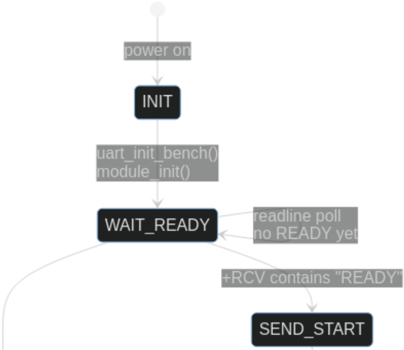
Macro	Value	Board	Notes
`HANDSHAKE_BEACON_MS`	1000	S3	Interval between READY beacons
`HANDSHAKE_LISTEN_MS`	500	S3	Listen window after each beacon for START
`HANDSHAKE_TIMEOUT_MS`	30000	DevKitV1	Max time to wait for READY before aborting
`HANDSHAKE_SETTLE_MS`	100	DevKitV1	Delay after sending START before first send

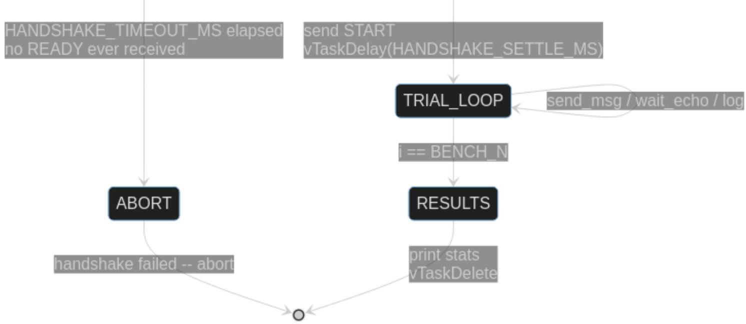
S3 handshake state machine


power on
INIT


uart_init_bench()
module_init()
BEACONING
send READY beacon
listen HANDSHAKE_LISTEN_MS
no START received
+RCV contains "START"
link confirmed
LISTENING
readline timeout (10 s)
line received
not +RCV= (ignore)
FILTER
parse_rcv() failed
delay 15 ms
send_echo()
wait +OK (500 ms)
starts with +RCV=
PARSE
src, len, data valid
ECHO

DevKitV1 handshake state machine


power on
INIT
uart_init_bench()
module_init()
WAIT_READY
readline poll
no READY yet
+RCV contains "READY"
SEND_START


HANDSHAKE_TIMEOUT_MS elapsed
no READY ever received
send START
vTaskDelay(HANDSHAKE_SETTLE_MS)
TRIAL_LOOP
send_msg / wait_echo / log
i == BENCH_N
ABORT
RESULTS
handshake failed -- abort
print stats
vTaskDelete

Bug 5 – S3: System freeze on first RF transmission (TX power / power rail)

Observed behavior:

The S3 completes `module_init` (all AT configuration commands succeed), prints `waiting for handshake...`, enters `wait_for_start`, and sends its first `AT+SEND=2,5,READY`. The system immediately freezes – no further log output from any task (including the heap monitor running in `app_main`). No crash dump, no watchdog trace, no reboot. The serial monitor shows garbled characters at the point of freeze.

On rare occasions the LoRa module completes the RF transmission before the ESP32-S3 dies, and the DevKitV1 receives the READY packet – but only during the S3's death/reset event, never during normal operation.

Root cause:

The RYLR998 defaults to `AT+CRFOP=22` (maximum TX output power, +22 dBm). At this power level the SX1262 PA draws approximately **120 mA** during RF transmission. This current is drawn from the ESP32-S3 dev board's onboard 3.3 V LDO, which is also powering the ESP32-S3 itself (~100-200 mA). The combined load exceeds the regulator's capacity, causing the 3.3 V rail to sag.

The voltage drop is not deep enough to trigger the ESP32-S3's brownout detector (which would produce `rst:0x12 BROWNOUT_RESET`), but it is enough to corrupt the CPU state or stall the USB Serial JTAG debug interface. The result is a system that appears completely frozen – no tasks run, no interrupts fire, no output is produced.

All AT commands during `module_init` (`AT+RESET`, `AT+ADDRESS`, `AT+NETWORKID`, `AT+PARAMETER`, queries) succeed because they are **configuration-only** – they do not activate the RF power amplifier. `AT+SEND` is the first and only command that triggers actual RF transmission and the associated current spike.

Why this wasn't caught earlier:

The DevKitV1 (ESP32) uses a different dev board with a more capable LDO and does not exhibit the same power rail sag at CRFOP=22. The problem is specific to the ESP32-S3 board's power delivery.

Fix (software):

Reduce TX output power before any `AT+SEND` by adding `AT+CRFOP=7` to `module_init` on both boards. At +7 dBm (~5 mW), TX current drops to approximately 30 mA – well within the LDO's headroom. This is sufficient for bench testing at desk distances (RSSI ~-25 to -45 dBm, SNR ~13-14 dB observed).

```
/* In module_init(), after AT+PARAMETER, before verification queries */
at_cmd("AT+CRFOP=7\r\n", "+OK", LORA_AT_TIMEOUT_MS);
```

Fix (hardware, for higher TX power / field range):

The software fix limits range. To operate at higher CRFOP (needed for field distances of 1+ miles), the LoRa module's power supply must be decoupled from the ESP32's rail:

1. **Bulk capacitor (simplest PCB rework):** Solder a 100-470 uF electrolytic capacitor directly across the RYLR998's VCC and GND pins. The capacitor acts as a local energy reservoir – it stores charge between transmissions and releases it instantly during the TX current spike, preventing the voltage rail from sagging while the regulator catches up.

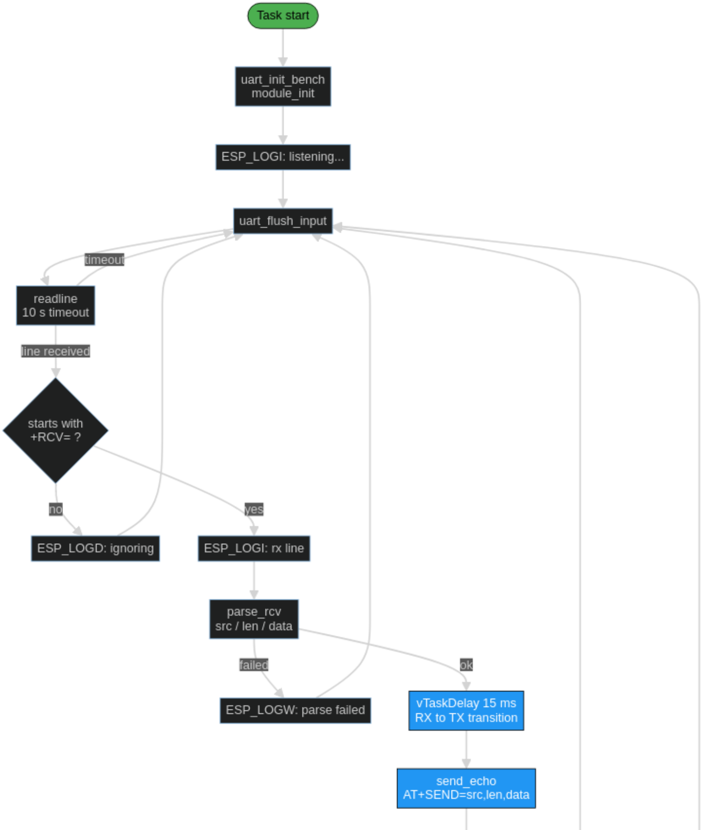
2. **Separate regulator (next board revision):** Feed the RYLR998 from a dedicated 3.3 V LDO (e.g., AMS1117-3.3) powered from the 5 V USB rail. This fully isolates the module's TX transients from the ESP32's power domain.

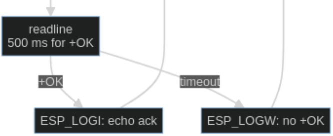
CRFOP vs range reference:

CRFOP	TX Power	Approx. TX Current	Range (LOS)	Notes
7	+7 dBm	~30 mA	~200 m	Safe on dev board LDO
14	+14 dBm	~60 mA	~1 km	May need cap
22	+22 dBm	~120 mA	~5+ km	Requires cap or separate regulator

Corrected Data Flow

S3 echo slave – full corrected loop


Task start
uart_init_bench
module_init
ESP_LOGI: listening...
uart_flush_input
timeout
readline
10 s timeout
line received
starts with
+RCV= ?
no
yes
ESP_LOGD: ignoring
ESP_LOGI: rx line
parse_rcv
src / len / data
failed
ok
ESP_LOGW: parse failed
vTaskDelay 15 ms
RX to TX transition
send_echo
AT+SEND=src,len,data


readline
500 ms for +OK
+OK
timeout
ESP_LOGI: echo ack
ESP_LOGW: no +OK

Configuration Reference

Both `config.h` files must define matching RF parameters. Bench mode is gated by `#define TEST_MODE_LORA_BENCH` on both boards.

Macro	DevKitV1	(WROOM) S3	Notes
`BENCH_SF`	7	7	Spreading factor – **must match**
`BENCH_BW`	9	9	500 kHz – **must match**
`BENCH_CR`	1	1	Coding rate 4/5 – **must match**
`BENCH_PREAMBLE`	12	12	Preamble symbols – **must match**
`LORA_NETWORK_ID`	6	6	Network ID – **must match**
`LORA_ADDRESS_SENDER`	2	–	DevKitV1 module address
`LORA_ADDRESS_RECEIVER`	–	1	S3 module address
`BENCH_DEST_ADDR`	1	–	DevKitV1 sends to S3 addr 1
`LORA_UART_NUM`	UART_NUM_2	UART_NUM_2	Different peripheral per board
`LORA_PIN_TX`	GPIO17	GPIO17	ESP32 TX -> Module RX
`LORA_PIN_RX`	GPIO16	GPIO18	Module TX -> ESP32 RX
`CRFOP`	7	7	TX power (dBm) – see Bug 5
`BENCH_PKT_SIZE`	16	–	Bytes per packet
`BENCH_N`	50	–	Packets per trial
`BENCH_TIMEOUT_MS`	10000	–	Per-packet echo timeout (ms)
`BENCH_INTER_PACKET_MS`	10	–	Inter-packet gap (ms)
`LORA_AT_TIMEOUT_MS`	500	500	AT command response timeout (ms)

Expected bench results (bench distance)

Metric	Expected
Delivery	50 / 50 (100%)
Avg RTT	~160 ms
RSSI	~-44 dBm
SNR	~13 dB

References

- SDD003 – LoRa Link Budget & Spreading Factor Calculations
- REYAX RYLR998 AT Command Reference
- ESP-IDF UART Driver API

Source files modified by this SDD

File	Change
`esp32s3_wroom/main/src/lora_bench.c`	Remove UART reinit; add 15 ms delay; add READY beaco
`esp32s3_wroom/main/inc/config.h`	Change LORA_UART_NUM to UA
`esp32_devkitv1/main/src/interprocessor/lora_bench.c`	Fix null byte; fix time units; simplify size clamp; add READY/S
`esp32_devkitv1/main/inc/config.h`	Add HANDSHAKE_TIMEOUT_MS, HANDS
`esp32s3_wroom/main/inc/config.h`	Add HANDSHAKE_BEACON_MS, HANDS